A Deformation for Homotopy Fixed Points Spectral Sequences

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Synthetic spectra

In [1], Pstrągowski introduces the ∞-category of *E-synthetic spectra* $\mathcal{S}yn_E$ for a homotopy ring spectrum E of Adams-type.

In particular, $\mathcal{S}yn_E$ categorifies the *E-based Adams spectral sequences* in $\mathcal{S}p$, providing useful computational techniques for Adams spectral sequences.

Goal: To generalize the construction of synthetic spectra to equivariant homotopy theory, aiming to categorify the *homotopy fixed points spectral sequences*.

Synthetic spectra, continued

The category of E-synthetic spectra can be interpreted in two ways:

Definition ([1], Pstrągowski, 2023)

The ∞-category of E-synthetic spectra

$$\mathcal{S}yn_E \simeq Sh_{\Sigma}^{\mathcal{S}p}(\mathcal{S}p_E^{\mathrm{fp}})$$

is defined as the category of spherical sheaves of spectra on the full subcategory of finite E_*-projective spectra.

Theorem ([2],Lawson, 2024)

Whenever E is *connective*, there is a symmetric monoidal equivalence

$$\mathcal{S}yn_E \simeq \mathcal{M}od_{\mathrm{Déc}\ AF(\mathbb{S})}(\mathrm{Fil}(\mathcal{S}p)),$$

where Déc $AF(\mathbb{S})$ is a certain commutative filtered ring spectrum.

Equivariant generalization

Along the filtered object viewpoint, in [3], Angelini-Knoll, Behrens, Belmont, and Kong introduce the category of *complete artificial motivic spectra*

$$\widehat{\mathcal{SH}}^{\mathrm{art}}_{G,p} \simeq \mathcal{M}od_{\mathbb{S}^{\mathrm{def}}_{G,p}}(\mathrm{Fil}(\mathcal{S}p^{BG}_{\mathrm{ip}})).$$

This categorifies *the modified Adams-Novikov spectral sequences* for Borel G-spectra.

In this talk, we will focus on the sheaf-theoretical viewpoint. We construct the category of $\mathbb{S}[G]$-synthetic spectra $\mathcal{S}yn_{\mathbb{S}[G]}$ which categorifies the *homotopy fixed point spectral sequences* in the ∞-category of genuine G-spectra $\mathcal{S}p_G$.

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∞-categorical deformation theory

We begin with some general features of the ∞-categorical deformation theory.
Let $\mathrm{Fil}(\mathcal{S}p) \simeq \mathrm{Fun}((\mathbb{Z}^{\leqslant})^{\mathrm{op}}, \mathcal{S}p) \in \mathrm{CAlg}(\mathcal{P}r^{\mathrm{L}}_{\mathrm{st}})$ be the category of filtered spectra.

Definition ([4], Barkan, 2023)

A *one-parameter deformation* is a $\mathrm{Fil}(\mathcal{S}p)$-module $\mathcal{C}^{\mathrm{def}}$ in $\mathcal{P}r^{\mathrm{L}}_{\mathrm{st}}$. Moreover, given an ∞-operad $\mathcal{O}$, an *$\mathcal{O}$-monoidal one-parameter deformation* is an $\mathcal{O}$-algebra in $\mathcal{M}od_{\mathrm{Fil}(\mathcal{S}p)}(\mathcal{P}r^{\mathrm{L}}_{\mathrm{st}})$.
The *generic fiber* of $\mathcal{C}^{\mathrm{def}}$ is the base-change along the underlying spectra functor

$$\mathcal{S}p \otimes_{\mathrm{Fil}(\mathcal{S}p)} \mathcal{C}^{\mathrm{def}} \in \mathcal{M}od_{\mathcal{S}p}(\mathcal{P}r^{\mathrm{L}}_{\mathrm{st}}) \simeq \mathcal{P}r^{\mathrm{L}}_{\mathrm{st}}.$$

The *special fiber* of $\mathcal{C}^{\mathrm{def}}$ is the base-change along the associated graded functor

$$\mathrm{Gr}(\mathcal{S}p) \otimes_{\mathrm{Fil}(\mathcal{S}p)} \mathcal{C}^{\mathrm{def}} \in \mathcal{M}od_{\mathrm{Gr}(\mathcal{S}p)}(\mathcal{P}r^{\mathrm{L}}_{\mathrm{st}}).$$

$\mathrm{Fil}(\mathcal{S}p)$-module structure

The following structures of $\mathcal{C}^{\mathrm{def}}$ can be deduced from the $\mathrm{Fil}(\mathcal{S}p)$ action:

- $\mathrm{Fil}(\mathcal{S}p)$-*enrichment*: $\mathcal{C}^{\mathrm{def}}$ is canonically enriched in $\mathrm{Fil}(\mathcal{S}p)$;
- *Bi-grading*:
$$\Sigma^{a,b} : \mathcal{C}^{\mathrm{def}} \to \mathcal{C}^{\mathrm{def}};$$
- *The deformation parameter* τ: a natural transformation
$$\tau : \Sigma^{0,-1} \to id_{\mathcal{C}^{\mathrm{def}}};$$
- *The generic fiber* coincides with the localization $\mathcal{C}^{\mathrm{def}}[\tau_X^{-1}]$;
- *The special fiber* coincides with the category of modules $\mathcal{M}od_{C\tau}(\mathcal{C}^{\mathrm{def}})$ over a certain monad $C\tau$ on $\mathcal{C}^{\mathrm{def}}$.

Spectral sequences

As desired, we expect a deformation $\mathcal{C}^{\mathrm{def}}$ would categorify a certain type of spectral sequence defined in its generic fiber $\mathcal{C} := \mathcal{S}p \otimes_{\mathrm{Fil}(\mathcal{S}p)} \mathcal{C}^{\mathrm{def}}$.

To see this, let $\nu : \mathcal{C} \to \mathcal{C}^{\mathrm{def}}$ be a suitable section (merely a functor between ∞-categories) of $\tau^{-1} : \mathcal{C}^{\mathrm{def}} \to \mathcal{C}$.

Then, for each pair of objects $X, Y \in \mathcal{C}$, the $\mathrm{Fil}(\mathcal{S}p)$-valued internal hom

$$\mathsf{F}_*(\nu(X), \nu(Y)) \in \mathrm{Fil}(\mathcal{S}p)$$

defines a filtered spectrum whose underlying spectrum is $\mathsf{F}(X, Y) \in \mathcal{S}p$, which further determines a spectral sequence.

Examples of deformations

Example

For any $\mathcal{C} \in \mathcal{P}r^{\mathrm{L}}_{\mathrm{st}}$, the ∞-category of *filtered objects* in $\mathcal{C}$

$$\mathrm{Fil}(\mathcal{C}) \simeq \mathrm{Fil}(\mathcal{S}p) \otimes \mathcal{C}$$

is a deformation with the generic fiber $\mathcal{C}$ and the special fiber $\mathrm{Gr}(\mathcal{C})$. This is also the free $\mathrm{Fil}(\mathcal{S}p)$-module on $\mathcal{C}$.

Example

Let E be a homotopy ring spectrum of Adams-type. Then the category of *E-synthetic spectra* $\mathcal{S}yn_E$ is a deformation with the generic fiber $\mathcal{S}p$.

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A sheaf-theoretical construction

Now, we modify the construction of synthetic spectra and introduce a sheaf-theoretical machine that produces deformation categories of well-behaved stable ∞-categories.

To be more precise, we assume that the ground category $\mathcal{C}$ satisfies the following condition:

A sheaf-theoretical construction, continued

Assumptions A

Let $\mathcal{C} \in \mathrm{CAlg}(\mathcal{P}r^{\mathrm{L}}_{\mathrm{st}})$ be a rigidly generated, presentably symmetric monoidal stable ∞-category. Let $\tilde{\mathcal{C}}$ be a full subcategory of $\mathcal{C}$ such that

1. all elements in $\tilde{\mathcal{C}}$ are compact in $\mathcal{C}$;
2. the zero object $0 \in \mathcal{C}$ in contained in $\tilde{\mathcal{C}}$ and $\tilde{\mathcal{C}}$ is closed under finite products, suspensions, and desuspensions;
3. the unit $\mathbb{1}_{\mathcal{C}} \in \mathcal{C}$ in contained in $\tilde{\mathcal{C}}$ and $\tilde{\mathcal{C}}$ is closed tensor product;
4. $\mathcal{C}$ is generated by $\tilde{\mathcal{C}}$ under all colimits.

Moreover, let $\mathcal{Q} \subset \mathrm{Fun}(\Delta^1, \tilde{\mathcal{C}})$ be a class of morphisms in $\tilde{\mathcal{C}}$ such that

1. all equivalences are in $\mathcal{Q}$;
2. $\mathcal{Q}$ is closed under composites;
3. given any $x \to y$ in $\mathcal{Q}$ and an arbitrary map $z \to y$, their pull-back (computed in $\mathcal{C}$) $x \times_y z$ is contained in $\tilde{\mathcal{C}}$ and $x \times_y z \to z$ is in $\mathcal{Q}$;
4. $\mathcal{Q}$ is closed under tensoring with any $x \in \tilde{\mathcal{C}}$.

A sheaf-theoretical construction, continued

Theorem

Under Assumptions A, $\mathcal{Q}$ determines a Grothendieck pretopology on $\tilde{\mathcal{C}}$, and the category of spherical sheaves of spectra

$$\mathrm{Def}(\mathcal{C}, \mathcal{Q}) := Sh^{\mathcal{S}p}_{\Sigma}(\tilde{\mathcal{C}})$$

admits a structure of being an $\mathbb{E}_\infty$-monoidal one-parameter deformation. Moreover, the generic fiber is equivalent (as $\mathcal{S}p$-algebras) to $\mathcal{C}$; that is,

$$\mathcal{S}p \otimes_{\mathrm{Fil}(\mathcal{S}p)} \mathrm{Def}(\mathcal{C}, \mathcal{Q}) \simeq \mathcal{C} \in \mathsf{CAlg}(\mathcal{P}r^{\mathsf{L}}_{\mathsf{st}}).$$

The special fiber admits a well-behaved t-structure such that its left separated part is equivalent (as $\mathrm{Gr}(\mathcal{S}p)$-algebras) to the (unbounded) derived ∞-category of its heart; that is,

$$\mathcal{M}od^{\mathrm{sep}}_{\mathcal{C}^\tau}(\mathrm{Def}(\mathcal{C}, \mathcal{Q})) \simeq \mathcal{D}(Sh^{\mathcal{S}et}_{\Sigma}(\tilde{\mathcal{C}})) \in \mathsf{CAlg}_{\mathrm{Gr}(\mathcal{S}p)}(\mathcal{P}r^{\mathsf{L}}_{\mathsf{st}})$$

The synthetic analogue

As mentioned above, in order to compare the deformation category with spectral sequences, we introduce a well-behaved section of

$$\tau^{-1} : \mathrm{Def}(\mathcal{C}, \mathcal{Q}) \to \mathcal{C}$$

called the *synthetic analogue*.

Definition

We define the *synthetic analogue* functor $\nu : \mathcal{C} \to \mathrm{Def}(\mathcal{C}, \mathcal{Q})$ to be the following composite:

$$\mathcal{C} \xrightarrow{y} P_{\Sigma}(\tilde{\mathcal{C}}) \xrightarrow{L} Sh_{\Sigma}(\tilde{\mathcal{C}}) \xrightarrow{\Sigma^{\infty}_{+}} Sh^{\mathcal{S}p}_{\Sigma}(\tilde{\mathcal{C}}) \simeq \mathrm{Def}(\mathcal{C}, \mathcal{Q}),$$

where y is the Yoneda embedding and L is the sheafification.

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$\mathbb{S}[G]$-synthetic spectra

Throughout this section, we fix a *finite* group G and let $\mathcal{S}p_G$ be the ∞-category of genuine G-spectra.
Let $\mathbb{S}[G] \in \mathrm{CAlg}(\mathcal{S}p_G)$ be the group ring.

Proposition

The category $\mathcal{S}p_G$, the full subcategory $\mathcal{S}p_G^\omega$ spanned by compact G-spectra, the collection $\mathcal{Q}$ of $\mathbb{S}[G]$-split morphisms in $\mathcal{S}p_G^\omega$ satisfies the conditions in Assumptions A. Here, we say a morphism $f : X \to Y$ in $\mathcal{S}p_G^\omega$ is $\mathbb{S}[G]$-*split* if $f \otimes \mathbb{S}[G]$ admits a section.

Definition

We define the ∞-category $\mathcal{S}yn_{\mathbb{S}[G]}$ of $\mathbb{S}[G]$-*synthetic spectra* as the deformation category $\mathrm{Def}(\mathcal{S}p_G, \mathcal{Q}) \simeq Sh_\Sigma^{\mathcal{S}p}(\mathcal{S}p_G^\omega)$.

Homotopy fixed point spectral sequences

Our next goal is to explain the relation between $\mathcal{S}yn_{\mathbb{S}[G]}$ and the homotopy fixed point spectral sequences.

We begin with the following fact.

Proposition ([5], Mathew-Naumann-Noel, 2017)

For any $X \in \mathcal{S}p_G$, the ($\mathbb{Z}$-graded, Mackey-functor-valued) homotopy fixed point spectral sequence of X is isomorphic to the $\mathbb{S}[G]$-Adams spectral sequence of X.

Therefore, using the standard construction of Adams spectral sequences, we introduce the following definition.

Homotopy fixed point spectral sequences, continued

Let $X \in \mathcal{S}p_G$ be any G-spectrum and let $\overline{\mathbb{S}[G]}$ be the fiber of the unit

$$\mathbb{S} \to \mathbb{S}[G].$$

We define the $\mathbb{S}[G]$-*Adams filtration* $AF_*(X) \in \mathrm{Fil}(\mathcal{S}p_G)$ of X as the following filtered G-spectrum

$$\cdots \to \overline{\mathbb{S}[G]}^{\otimes n+1} \otimes X \to \overline{\mathbb{S}[G]}^{\otimes n} \otimes X \to \cdots \to \overline{\mathbb{S}[G]} \otimes X \to X \to X \to \cdots .$$

More precisely,

$$AF_n(X) := \begin{cases} \overline{\mathbb{S}[G]}^{\otimes n} \otimes X, & \text{if } n > 0; \\ X, & \text{if } n \leqslant 0. \end{cases}$$

Comparison with HFPSS

In order to compare $\mathbb{S}[G]$-synthetic spectra with filtered G-spectra, we need to construct a symmetric monoidal left adjoint $c : \mathcal{S}p_G \to \mathcal{S}yn_{\mathbb{S}[G]}$.

Proposition

There exists a unique symmetric monoidal left adjoint

$$c : \mathcal{S}p_G \to \mathcal{S}yn_{\mathbb{S}[G]}$$

such that

$$c \circ y \simeq \nu \circ y : \mathrm{Span}(\mathrm{Fin}_G) \xrightarrow{y} \mathcal{S}p_G \to \mathcal{S}yn_{\mathbb{S}[G]}$$

where y is the Yoneda embedding.
Moreover, c is a section of $\tau^{-1} : \mathcal{S}yn_{\mathbb{S}[G]} \to \mathcal{S}p_G$.

Comparison with HFPSS, continued

Now we explain the relation between $\mathcal{S}yn_{\mathbb{S}[G]}$ and the homotopy fixed point spectral sequences.

Theorem

By tensoring with $c : \mathcal{S}p_G \to \mathcal{S}yn_{\mathbb{S}[G]}$, we get an adjoint pair

$$\mathrm{Fil}(\mathcal{S}p_G) \simeq \mathrm{Fil}(\mathcal{S}p) \otimes \mathcal{S}p_G \underset{i_*}{\overset{i^*}{\rightleftarrows}} \mathcal{S}yn_{\mathbb{S}[G]}.$$

For any G-spectra $X \in \mathcal{S}p_G$, we have an equivalence

$$i_*(\nu X) \simeq \text{Déc } AF_*(X) \in \mathrm{Fil}(\mathcal{S}p_G).$$

Here, Déc $: \mathrm{Fil}(\mathcal{S}p_G) \to \mathrm{Fil}(\mathcal{S}p_G)$ is the *décalage* defined using the standard t-structure on $\mathcal{S}p_G$.

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Thank You!